Steady Strength: audit of the Master Training Library and Handover

Date: 3 Nov 2026. Reviewer: Claude (programming/data audit, **not** a qualified strength-and-conditioning or clinical review).
Everything below is labelled **VERIFIED** (I checked it myself against the PDFs), **PROPOSED** (my recommendation, needs your approval) or **NEEDS EXTERNAL VERIFICATION**.

0. What I could and could not check

Still needed from you to finish this properly: the 240-session .docx (authoritative for the perimenopause pathway) plus the three other PDFs. Also see `ADDENDUM_missing_deliverables.md` and `module_compatibility_matrix.csv`.

- Supplied: the Handover PDF (3 pp) and the Master Library and App Build Guide (43 pp).
- Not supplied: the source .docx and the other three PDFs the handover refers to. **I cannot confirm the master faithfully reflects them.**
- No exercise videos, licences or citation texts were supplied. Nothing below about video rights or research claims is verified.
- Draft data produced alongside this report (`catalogue_draft.json`) is parsed from the PDF text. It is marked DRAFT_UNVERIFIED and must not ship as production content.

1. Verified discrepancies (claim, finding, proposed fix)

#	Claim in document	Finding (VERIFIED)	Proposed resolution
1	Appendix A: "30 distinct session templates"	Only **21** are defined. F3 is never scheduled.	Correct the claim; drop or schedule F3.
	Page 21: "19	**19 is correct** for Appendix B, but	

2	session templates" (Appendix B)	this is a *different* set from Appendix A's 21.	Treat A and B as two systems; see #3.
2b	Appendix A is described as the "uploaded" pathways	It holds **five different programs** (Strong Foundations 2d, Stronger Every Week 3d, Strength & Shape 4d, Strength Plus 5d, The Strength Routine 6d; 24+36+48+60+72 = 240 rows), each 12 weeks with phases Find Your Strength (wk 1-4), Build Your Strength (5-8), Feel Stronger (9-12). Appendix B holds a **second, different set of five** (Strength Foundations, Build & Grow, Lower Body Focus, Muscle Builder Pro, Strong for Life). So the master contains **10 programs, not 5**, with near-identical names ("Strong Foundations" vs "Strength Foundations").	Pick one catalogue (section 3).
2c	Template names reused for different things	"Lower Body Strength" is template X1 in A and a lower specialisation in B; "Strength & Stability" is both S1 and X3; "Full Body Strong" is both F3 and S3.	Unique display names enforced by a validator.
2d	Acceptance test 1: "all 240 assignments resolve to a template **and week-specific prescription**"	Templates resolve, but A has **no sets/reps/RIR**, so week-specific prescriptions cannot exist. **Test 1 cannot pass as written.** Test 7 (weeks 4 and 8 "consolidation rules") conflicts with B's optional-week-4 and with A's 4-week phase blocks.	Rewrite tests 1 and 7 (change request).
3	Appendix A templates list exercises with alternatives but **no sets/reps**;	A's F1 "Total Body Strength" is not the same session as B's Strength Foundations Full Body A. The two are **not reconciled**.	B is the only prescriptive source. A stays reference-only until sets/reps are supplied. Do not import A as

	Appendix B has sets/reps		workouts.
4	Appendix A has 240 schedule rows	All 240 resolve to a defined template.	OK, but they point to the un-prescribed A templates (see #3).
5	Duplicate templates in A	Identical exercise lists: F1=S1, F2=S2=X3, F3=S3=X5, L1=X1, U1=X2, U2=X4.	Merge to one template each; keep a name alias.
6	Name collisions	"Full Body Strong", "Strength & Stability" and "Strong/Strength Foundations" are used for different things in different places.	Final names in section 3.
7	Master p27 (base) and p31 (lower modules) say deloads are **optional**; p39 (upper modules) schedules a **planned week 4 at 50-60% of sets**	Contradictory deload policy.	One policy: planned lower-fatigue week is a user-visible *suggestion* driven by readiness and performance, never silent.
8	Modules are "6 weeks"; base programs are "12 weeks"	Master never says which base weeks a module occupies, or whether lower modules replace the base program's lower sessions.	PROPOSED: module runs as a 6-week block that overlays the base program; base week pointer pauses. Needs coach decision.
9	Upper modules "replace overlapping work, not add"	**Not true in practice.** See section 5: net +10 to +20 sets/week and +10 to +30 min per receiving session.	Add a mandatory "fit to session time" trim step (section 5).

10	Session length target (earlier brief) 45-60 min; lower modules say 45-70 min	My estimate for lower modules is 45-52 min incl. warm-up/cool-down; upper days become 75-90 min once a module is added.	Make duration a hard constraint in the engine.
11	"ACSM Position Stand on Resistance Training (2026)"	I cannot verify a 2026 ACSM position stand exists.	**NEEDS EXTERNAL VERIFICATION** before it appears anywhere in the app.
12	Ambiguous exercise slots	13 names are unspecified (e.g. "Squat", "Calf raise", "Leg curl", "Shoulder press", "Lateral raise", "Cable row"); 3 slots name two exercises ("Pull-up or pulldown", "Weighted or assisted pull-up", "Flat barbell or dumbbell press").	Pick one default exercise per slot (section 6).
13	93 exercise names	3 exact duplicates (triceps pressdown x2, DB shoulder press x2, machine chest press x2) merge to **90 IDs**. Many near-duplicates remain (Squat / Back squat / High-bar squat; Seated row / Seated cable row / Cable row).	Canonical ID table in `catalogue_draft.json`.
14	No sets/reps for time-based moves in a common field	Plank, farmer carry use seconds; dead bug, Pallof use per side; lunges per leg.	Slot needs `unit` and `per` fields (done in draft).
15	Rest, per-exercise load increment, target RIR per slot	Not specified per exercise.	Draft derives rest from the general rule. **Increments left null**; coach must set (earlier default 2.5 kg upper / 5 kg lower is only a placeholder).

2. Coverage audit of the five base programs (VERIFIED counts, my muscle mapping)

Weekly direct hard sets (primary muscle credited in full; compounds therefore overlap, so glute and quad numbers are inflated). Full table: `volume_audit.csv`.

Program	Standout numbers	Gaps
Strength Foundations (3d)	quads 9, lats 9, glutes 14, chest 6, hamstrings 4	No direct biceps, triceps, rear delts, calves. Acceptable for beginners; say so in the UI.
Build & Grow (4d)	glutes 14, lats 12, quads 11, chest 9, calves 6	Arms thin (biceps 3, triceps 2).
Lower Body Focus (4d)	glutes 21, quads 11	No calves. Upper volume low (biceps 2, triceps 2, rear delts 2). Name does not say "plus modest upper".
Muscle Builder Pro (5d)	chest 13, lats 13, quads 13, biceps 7	Triceps 3, rear delts 3.
Strong for Life (2-3d)	glutes 11, quads 6	**Hamstrings 2, chest 3, side delts 2** on the 2-day version. Very low; fine for maintenance, not for "all levels" growth.

Estimated session length incl. 8 min warm-up and 5 min cool-down (my heuristic: about 3.2 min per compound set, 2.0 per isolation set): base sessions 47-63 min. Only Build & Grow Upper A/B (61, 63 min) run slightly over 60.

3. Proposed catalogue and names (PROPOSED, needs your approval)

Core pathways, 12 weeks each:

Master name	Proposed final name	Days	Level
Strength			

Foundations	**Strength Foundations** (keep)	3	Beginner
Build & Grow	**Build & Grow** (keep)	4	Beginner to intermediate
Lower Body Focus	**Lower Emphasis** (avoids clash with lower modules)	4	Intermediate
Muscle Builder Pro	**Muscle Builder** ("Pro" over-promises)	5	Intermediate to advanced
Strong for Life	**Strong for Life** (keep)	2-3	All levels

Specialisation modules, 6 weeks each, 2 sessions per week:

- Lower: Quad Builder, Glute Growth, Hamstring Builder, **Balanced Legs** (was Complete Leg Development), **Calves & Shins** (was Calf & Lower Leg Focus), **Heavy Legs** (was Lower Body Strength).
- Upper: Arm Builder, Back Builder, Chest Builder, Shoulder Builder (handover names).

CORRECTION after re-reading the handover: Appendix A is the 12-week perimenopause pathway (5 splits, 240 scheduled assignments) and the handover says to treat the original .docx as the *authoritative source for its own pathway*. The master PDF only carries exercise names and the schedule, not sets/reps/RPE, but the .docx may well contain them. So I do **not** recommend retiring or deferring it. PROPOSED: keep it as its own separate pathway family (working name **Strong Through Midlife**, symptom-aware options, no blanket loading rule), keep it out of the core-program list until reconciled, and **import it only after you upload the .docx** so prescriptions come from the authority, not from me. The 6-day The Strength Routine stays in scope but gets a volume/recovery review rather than being dropped. The phase labels (Find Your Strength 1-4, Build Your Strength 5-8, Feel Stronger 9-12) belong to this pathway only.

Your Claude-drafted **Beginner 12-Week (Options A-D)** overlaps the master directly: A=2 days, B=3 days, C=4 days, D=5 days. PROPOSED: ship the master's five as the catalogue; retire Options A-D as separate programs but keep their 28-exercise cue library and the Learn-phase (weeks 1-4 at 2 sets, RPE 6-7) as the beginner on-ramp rule. **Decision for Riki.**

Your five RW Fitness spreadsheet programs become a separate **Coach Series** (Full Body 3-Day, Upper/Lower 4-Day, Body Part Split 5-Day, 12-Week, Intermediate 18-Week). They keep your own content; the known source quirks (mislabelled sessions, duplicate "Legs & Abs", date-corrupted reps) still need your input before import.

4. Lower specialisations (VERIFIED parse)

Module	Direct sets per week	Session min (A / B, est.)
Quad Builder	quads 16, glutes 8, calves 6	46 / 49
Glute Growth	glutes 17, hamstrings 8, abductors 4	50 / 47
Hamstring Builder	hamstrings 16, glutes 12, calves 6	47 / 45
Balanced Legs	glutes 14, quads 10, hamstrings 7, calves 6	49 / 46
Calves & Shins	calves 14, tibialis 6	49 / 47
Heavy Legs	glutes 15, quads 14, hamstrings 8	52 / 52

Safety flags: Nordic curl (assisted) has a high injury/soreness profile, so it needs a review flag and an easy regression. Back squat 4x4-6, trap-bar deadlift 3x4-6 and front squat 3x6-8 in Heavy Legs are advanced lifts with no beginner gate in the master.

5. Upper-module integration test (VERIFIED arithmetic, my assumptions on what counts as "removable isolation")

Replacing base isolation work in the two receiving sessions, the module still adds sets:

Base + module	Net sets added per week	Extra minutes per receiving session
Muscle Builder + Arms	+10	about +10
Muscle Builder + Shoulders	+10	about +13
Build & Grow + Shoulders	+13	about +16
Build & Grow + Arms	+15	about +15
Build & Grow + Chest	+16	about +23
Build & Grow + Back	+17	about +26

Full-body bases + any module	+16 to +20	about +20 to +30

Build & Grow Upper B is already 63 min, so with a module it would reach roughly 80-90 min. **The "replace, don't add" rule only works if the app also trims or moves work.** PROPOSED engine rule: the module replaces all *accessory* slots for the target muscle in the receiving session, then trims the lowest-priority remaining accessory until the session fits the user's time limit, and shows exactly what was removed. Beginners on 3-day full-body bases should not be offered upper modules by default.

6. Default picks for ambiguous slots (PROPOSED, coach to confirm)

Squat -> back squat (high-bar); Calf raise -> standing in lower sessions A, seated in B; Leg curl -> seated; Chest press -> machine; Shoulder press -> dumbbell seated; Lateral raise -> dumbbell; Cable row / Seated row -> Seated cable row; Biceps curl -> dumbbell; Triceps extension -> cable overhead; Pull-up or pulldown -> lat pulldown (pull-up unlocked after assisted pull-up success); Weighted or assisted pull-up -> assisted pull-up; Flat barbell or dumbbell press -> dumbbell press.

7. Compliance and safety gaps against the handover

- The master has **no readiness check, no screening flow, no rule for returning after missed weeks, no pregnancy/postpartum gating, and no stop-and-seek-help wording** in the app flow. Added to the change request.
- Master says "perimenopause-aware: do not automatically prescribe lighter loads". Kept; no medical claims.
- No videos or licences are specified. All video fields default to "none approved" with a search-query fallback until you supply licensed or self-filmed clips.
- Research list names authors but no verifiable citations. Do not show evidence claims in the app until verified.

8. NEEDS EXTERNAL VERIFICATION (cannot be done by Claude)

1. Qualified S&C coach review of all 19 base templates, 10 modules, progression and rep/RIR targets.

2. Clinical review of perimenopause, pregnancy/postpartum, osteoporosis and injury pathways.
3. Australian privacy (APPs), consumer law, advertising/health-claim and app-store health-app obligations.
4. Video and image rights (wger, MuscleWiki, YouTube etc. unresolved).
5. Exact citations (including the "2026 ACSM position stand").
6. Dietitian review of calorie floors; confirm Butterfly Foundation number (1800 33 4673) is current.
7. Name and logo availability for "Steady Strength" (trademark, domain, App Store).

9. Decisions needed from Riki

1. Approve the proposed renames in section 3.
2. Retire Beginner Options A-D in favour of the master's five (keeping the cue library and Learn phase)?
3. Module policy: overlay with the base pointer paused (proposed), or replace the base lower sessions?
4. One deload policy (suggestion-based, proposed).
5. Per-exercise load increments (or accept 2.5 kg upper / 5 kg lower as placeholders until the coach signs off).
6. Which Coach Series files you want in the launch build, and answers on the spreadsheet quirks.